

The Effects of Ball Size and Weight on Pitching Kinematics and Kinetics in Young Baseball Players After a Week of Play

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Introduction: In Little League Baseball, athletes younger than 12 compete on smaller fields equipped with shorter base path and pitching distances, as well as depressed mounds in comparison to standardized fields [1], and as they develop there is a transition to adult-sized fields [2]. Although there are changes to the youth baseball field, there are no changes in baseball size and weight. From Little League to Major League Baseball, all balls weigh 5 ounces and have a circumference of 9 inches. Baseball pitching is a complex action that if not done correctly can lead to lasting injury to the shoulder and elbow [3]. Elbow torque is a metric that has been directly related to the weight of baseball and speed of the pitcher's arm [4]. Youth leagues allow players to begin pitching to each other around age 7 [1]. This early in player development, many athletes struggle to properly hold the ball due to its large size and weight. Understanding the mechanics of pitching for all ages and the potential effects of ball modification could lead to a reduction in injury occurrences in an athlete's career. Thus, the purpose of this study was to determine the effects of youth athletes throwing smaller, lighter, and smaller and lighter baseballs on elbow varus torque, shoulder distraction force, and throwing arm internal rotation velocity.

Materials and Methods: Subjects were obtained from the 2023 Wake Forest University Summer Baseball Camps. Each subject's parent gave consent, and the child gave informed assent to participate. Participants ranged from 7-9 years old and attended all days of camp. Campers from week one played with a normal, 5oz., 9inch baseball. Campers from week two played with a lighter, 4oz, baseball. Campers from week three played with a lighter, 4oz, baseball with a 5% reduced circumference. Campers from week four played with a lighter, 3oz, baseball. Upon arrival, the participant's height and weight were recorded along with how the individual gripped the baseball. The participants threw their assigned ball from a pitching mound to a target 46ft away for a total of 5 throws each on the first and last days of camp. All data was collected through the markerless motion capture system KinaTrax cameras operating at 300Hz. A TrackMan system was used in order to capture the velocity of the ball, and accuracy was noted. The data was evaluated through an analysis of covariance (ANCOVA) test that determined statistical differences in elbow varus torque, shoulder distraction force, and internal rotation of throwing arm velocity between baseball modifications, while adjusting for baseline measurements. Tukey Post Hoc testing further investigated differences between ball modifications. All testing was performed in MATLAB.

Results and Discussion: In the first week of camp, data from 25 youth athletes was collected and 35 in the second week. Data collection has not yet been completed due to timing of the baseball camps. However, when completed, the ANCOVA test will determine differences in elbow varus torque, throwing arm rotational velocity, and shoulder distraction force between ball modification groups from the beginning to the end of each week.

Conclusion: This study aimed to address the effects of ball modifications on youth baseball athletes. As elbow torque is directly to baseball weight, the question of whether or not it is safe for young athletes to

participate in weighted ball drills is undetermined. Conclusions cannot be made until data analysis is complete.

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